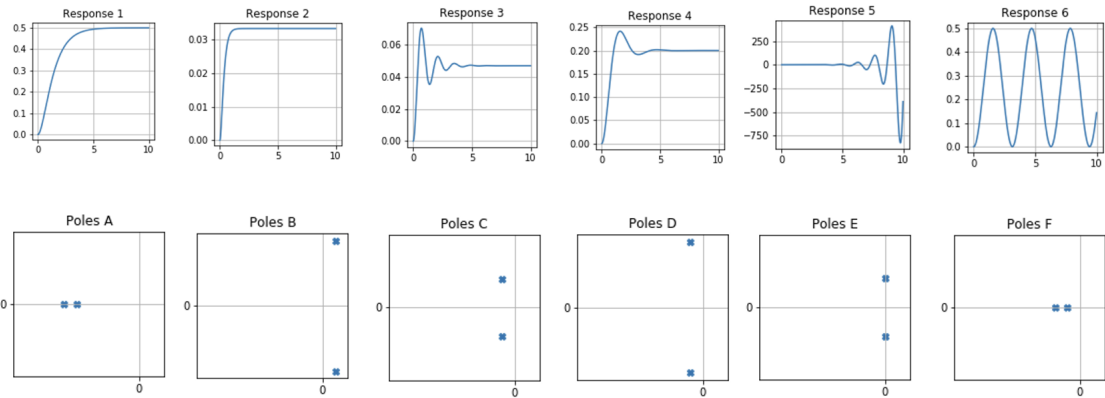


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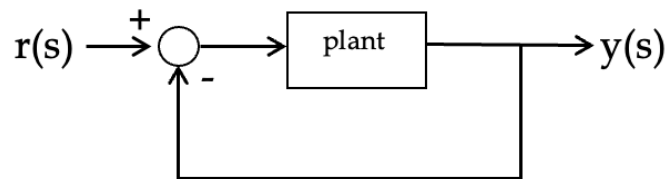
Problem Set 2.1 – Transfer Functions and System I.D.

Question 1

Each of the following plots 1–6 represents the **impulse response** for a second-order system, while the plots A–F represent their **pole locations**. Match each response to the corresponding pole plot.



Question 2



Each of the following

1. $G_1(s) = \frac{1}{s+2}$
2. $G_2(s) = \frac{s+1}{(s+5)(s+3)}$
3. $G_3(s) = \frac{1}{s(s+4)}$

is placed in closed loop, according to the diagram above. For each:

- (a) Calculate the **closed-loop transfer function**.

- (b) What is the **type number** of the system? Based on this, state whether you expect the system to track a unit step without error, and calculate the error constant for this input. If the error is non-zero, suggest a **modification** to the system that would lead to perfect tracking.
- (c) For the **open- and closed-loop** transfer functions, evaluate the **gain** of the system, and find the **initial** and **final** values of the unit-step response (use the initial- and final-value theorems where applicable).

Solutions

1. Calculate the closed-loop transfer function

For unity feedback with open-loop transfer function $Q(s)$, the closed-loop transfer function is

$$G_{CL}(s) = \frac{Q(s)}{1 + Q(s)}.$$

Writing $Q(s) = \frac{N_Q}{D_Q}$,

$$G_{CL}(s) = \frac{N_Q/D_Q}{1 + N_Q/D_Q} = \frac{N_Q}{D_Q + N_Q}.$$

Hence, for the given systems:

$$Q_1(s) = \frac{1}{s+2} \quad \Rightarrow \quad G_{CL,1}(s) = \frac{1}{s+3},$$

$$\begin{aligned} Q_2(s) &= \frac{s+1}{(s+5)(s+3)} \quad \Rightarrow \quad G_{CL,2}(s) = \frac{s+1}{s^2+8s+15+s+1} \\ &= \frac{s+1}{s^2+9s+16}, \end{aligned}$$

$$Q_3(s) = \frac{1}{s(s+4)} \quad \Rightarrow \quad G_{CL,3}(s) = \frac{1}{s(s+4)+1} = \frac{1}{s^2+s+1}.$$

2. Type number, step tracking, and error constant

- (a) For $Q_1(s) = \frac{1}{s+2}$: type 0 (no pole at the origin). The error transfer function is

$$\frac{1}{1 + Q_1(s)} = \frac{s+2}{s+3}.$$

For a unit step input, $E(s) = \frac{1}{s} \frac{s+2}{s+3}$. By the final value theorem,

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{s+2}{s+3} = \frac{2}{3}.$$

To achieve zero steady-state error to a step, add an integrator K/s in the forward path (make the system type 1).

(b) For $Q_2(s) = \frac{s+1}{(s+5)(s+3)}$: type 0.

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s^2 + 8s + 15}{s^2 + 9s + 16} = \frac{15}{16}.$$

(Again, adding an integrator in the forward path yields zero step error.)

(c) For $Q_3(s) = \frac{1}{s(s+4)}$: type 1 (one pole at the origin). The system tracks a step with zero steady-state error, so $e_{ss} = 0$.

3. Open- and closed-loop gains; initial and final values for a unit step response

Use the initial and final value theorems with input $1/s$.

System 1: $Q_1(s) = \frac{1}{s+2}$ *Open loop:*

$$\text{Gain} = \lim_{s \rightarrow 0} \frac{1}{s+2} = \frac{1}{2}, \quad y(0) = \lim_{s \rightarrow \infty} s \left(\frac{1}{s} \frac{1}{s+2} \right) = 0,$$

$$y(\infty) = \lim_{s \rightarrow 0} s \left(\frac{1}{s} \frac{1}{s+2} \right) = \frac{1}{2}.$$

Closed loop: $G_{CL,1}(s) = \frac{1}{s+3}$

$$\text{Gain} = \frac{1}{3}, \quad y(0) = 0, \quad y(\infty) = \frac{1}{3}.$$

System 2: $Q_2(s) = \frac{s+1}{(s+5)(s+3)}$ *Open loop:*

$$\text{Gain} = \lim_{s \rightarrow 0} \frac{s+1}{(s+5)(s+3)} = \frac{1}{15}, \quad y(0) = 0, \quad y(\infty) = \frac{1}{15}.$$

$$\text{Closed loop: } G_{CL,2}(s) = \frac{s+1}{s^2+9s+16}$$

$$\text{Gain} = \lim_{s \rightarrow 0} \frac{s+1}{s^2+9s+16} = \frac{1}{16}, \quad y(0) = 0, \quad y(\infty) = \frac{1}{16}.$$

$$\text{System 3: } Q_3(s) = \frac{1}{s(s+4)} \quad \text{Open loop:}$$

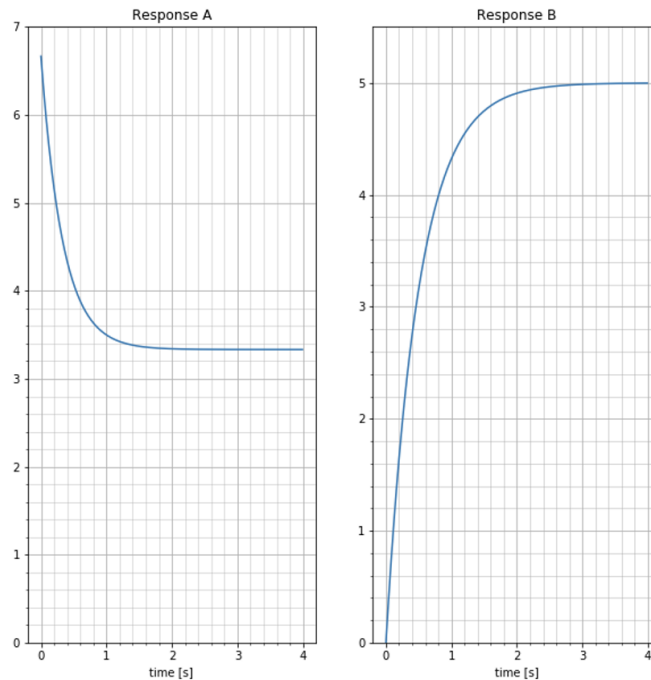
$$\text{Gain} = \lim_{s \rightarrow 0} \frac{1}{s(s+4)} = \infty, \quad y(0) = 0, \quad y(\infty) = \infty.$$

$$\text{Closed loop: } G_{CL,3}(s) = \frac{1}{s^2+s+1}$$

$$\text{Gain} = \lim_{s \rightarrow 0} \frac{1}{s^2+s+1} = 1, \quad y(0) = 0, \quad y(\infty) = 1.$$

Question 3

The plots below show the **step responses** obtained from two unknown **first-order** systems. Suggest transfer functions for the systems based on these plots.

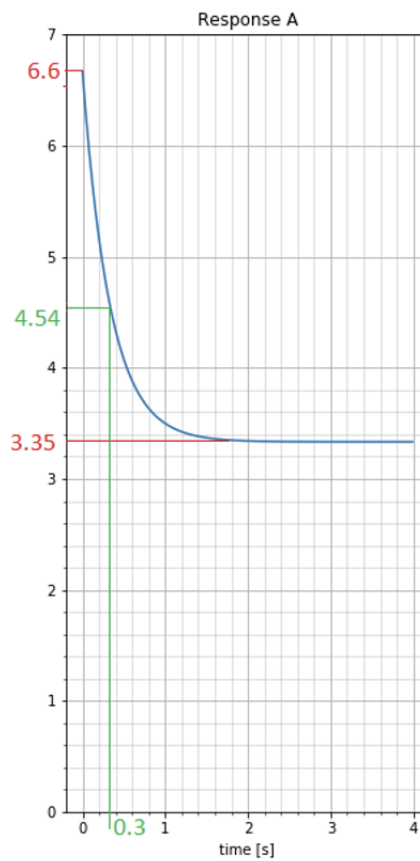


System Identification

A) The general form of the first-order transfer function we will use for system identification is:

$$\frac{A}{\tau s + 1}$$

Response A



Response A has a nonzero initial condition, so there must be a zero in the transfer function. We therefore modify the model to:

$$\frac{A \left(\frac{1}{z} s + 1 \right)}{\tau s + 1}$$

Gain: From the plot, the final value is approximately 3.35. Since the input step has height 1, the gain is:

$$A = 3.35$$

Time constant: For a first-order system, the time constant τ is the time at which the output reaches 63% of the total change. The initial value is ≈ 6.6 , so the change from initial to final value is:

$$6.6 - 3.35 = 3.25$$

The 63% point is:

$$6.6 - 0.63 \times 3.25 = 4.54$$

From the plot, this occurs at $t \approx 0.3$ s, so:

$$\tau = 0.3$$

Zero: Substituting known values into the transfer function:

$$\frac{3.35 \left(\frac{1}{z}s + 1 \right)}{0.3s + 1}$$

Applying the initial value theorem to the step response:

$$\lim_{s \rightarrow \infty} s \cdot \frac{1}{s} \cdot \frac{3.35 \left(\frac{1}{z}s + 1 \right)}{0.3s + 1} = \frac{3.35 \cdot \frac{1}{z}}{0.3}$$

Given the initial value 6.6:

$$\frac{3.35 \cdot \frac{1}{z}}{0.3} = 6.6$$

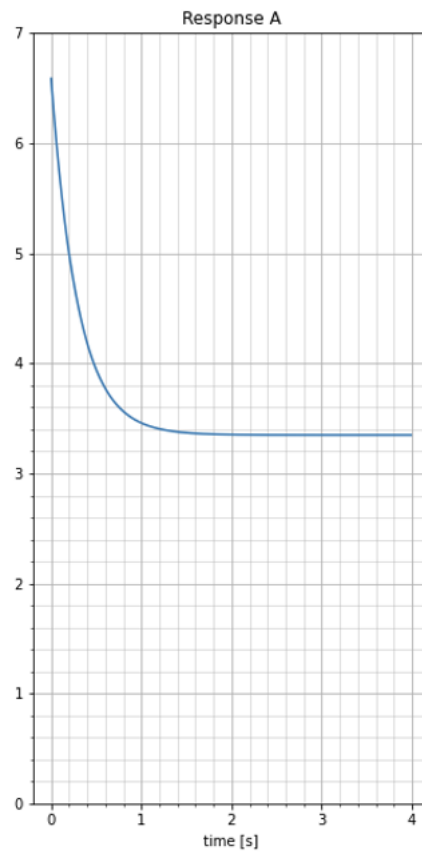
$$\frac{1}{z} = \frac{6.6 \times 0.3}{3.35} \approx 0.59$$

Final transfer function:

$$\frac{3.35 (0.59s + 1)}{0.3s + 1}$$

Simulating this transfer function gives a close match to the measured response.

I accidentally saved over the code I used to generate these plots, so I can't confirm the exact original function, but I simulated the transfer function and it looks about below.



Response B

This case is more straightforward because there is no zero.

Gain: Final value is 5, so:

$$A = 5$$

Time constant: 63% of the final value is:

$$0.63 \times 5 = 3.15$$

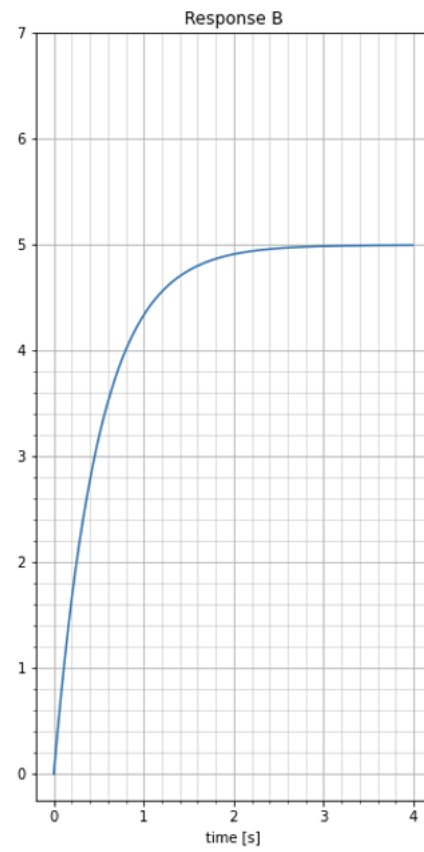
From the plot, this occurs at $t \approx 0.5$ s, so:

$$\tau = 0.5$$

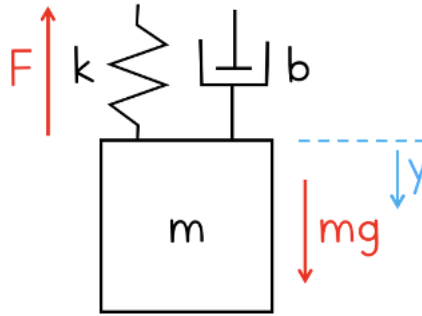
Final transfer function:

$$\frac{5}{0.5s + 1}$$

Simulation results confirm the correctness of this identification (below).



Question 4



A dumbwaiter is a small lift for carrying goods (usually food) between floors in a building. A force diagram is shown to the right. The input to the system is the motor force F , and the output is the vertical position y . The mass of the empty lift is $m = 5$ kg, and it can carry a maximum of 45 kg. The cables have spring constant $k = 100$ N/m. A damper with coefficient b moderates the speed. *Hint: pay attention to directions. You only need to consider downward motion.*

- Derive a transfer function relating **position** to **force**, $G(s) = \frac{Y(s)}{F(s)}$. Draw a block diagram depicting the force, position, gravity, and plant.
- It is vital that the system does **not oscillate**, so food does not get spilled. What is the **smallest damping coefficient** that would prevent oscillation, regardless of how full the lift is?
- Using this damping value, **plot** the response of the system to a **unit-step** input in force.
- How would you **modify the system** to track position setpoints accurately?
- How would you modify the system to **counteract gravity**?
- How would you modify the system so the user could specify a constant **velocity** for it to travel at?

Method

- (a) **Evaluate a transfer function for this system, relating the position and force. Draw a block diagram depicting the force, position, gravity, and plant.**

Taking downward as the positive direction, the equation of motion for the lift is:

$$m\ddot{y}(t) = -b\dot{y}(t) - ky(t) - F(t) + mg u(t)$$

(Since we're assuming right-sided signals and the weight is a constant value, we can model it as a step.)

Assuming initial rest, we can transform this to the s -plane using:

$$\frac{d^n}{dt^n}x(t) \longrightarrow s^n X(s)$$

$$ms^2Y(s) = -bsY(s) - kY(s) - F(s) + \frac{mg}{s}$$

Rearranging:

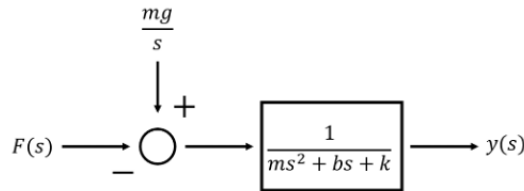
$$Y(s) = -\frac{1}{ms^2 + bs + k}F(s) + \frac{mg}{s} \cdot \frac{1}{ms^2 + bs + k}$$

So you can think of the output as a combination of a response to the input and a response to the weight.

The transfer function relating the force input and output is:

$$-\frac{1}{ms^2 + bs + k}$$

We can model the effect of gravity as an input disturbance.



- (b) **It is vital that this system does not oscillate, so food doesn't get spilled. What is the smallest damping coefficient that would prevent oscillation, regardless of how full the lift is?**

The characteristic equation is:

$$ms^2 + bs + k = 0$$

The poles are located at:

$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

For strictly non-oscillatory response (over- or critically damped), the poles must be real:

$$b^2 - 4mk \geq 0 \Rightarrow b \geq \sqrt{4mk}$$

The worst-case mass is $m = 50$ kg, so:

$$b \geq \sqrt{4(50)(100)} \Rightarrow b \geq 141.2$$

- (c) Using this damping value, plot the response of the system to a unit step input.

Let $m = 5$ kg, giving the transfer function:

$$\frac{1}{5s^2 + 141.2s + 100}$$

This can be rewritten as:

$$\frac{1}{5} \cdot \frac{1}{s^2 + 28.28s + 20}$$

Poles are at:

$$-0.73 \quad \text{and} \quad -27.56$$

The distance between the poles is 26.83, giving residuals of $\pm \frac{1}{26.83} \approx \pm 0.037$. Dividing by 5 (due to the $\frac{1}{5}$ factor) gives:

$$-\frac{0.0007}{s + 27.56} + \frac{0.0007}{s + 0.73}$$

Inverse Laplace transform (impulse response):

$$-0.0007 e^{-27.56t} + 0.0007 e^{-0.73t}$$

Step response (integral in time domain):

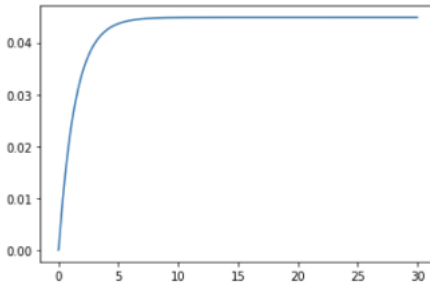
$$\frac{0.0007}{27.56} (e^{-27.56t} - 1) u(t) - \frac{0.0007}{0.73} (e^{-0.73t} - 1) u(t)$$

The combined input is $(mg - 1)u(t)$, with:

$$5 \times 9.81 - 1 = 48.05$$

So the scaled step response is:

$$48.05 \cdot \frac{0.0007}{27.56} (e^{-27.56t} - 1) u(t) - 48.05 \cdot \frac{0.0007}{0.73} (e^{-0.73t} - 1) u(t)$$



(d) **How to modify the system for various objectives:**

- (a) **Accurate position tracking:** Step 1: Put it in feedback. Step 2: Add an integral controller to increase the type number from 0 to 1.
- (b) **Counteract gravity:** Adding integral control in the feedback loop will reject the constant disturbance caused by gravity.
- (c) **Constant velocity tracking:** Constant velocity = ramp input (second-order). With feedback, add a second integrator to increase the type number to 2.

Question 5

For each of the following third-order systems,

$$1. G_a(s) = \frac{1}{(s+1)(s+10)(s+12)}$$

$$2. G_b(s) = \frac{s+2.1}{(s+2)(s+5)(s+10)}$$

$$3. G_c(s) = \frac{1}{(s+5)(s+5.5)(s+7)}$$

- (a) Decompose the system into its **modes** and calculate the **residuals** (partial-fraction expansion) for each mode.
- (b) Suggest **second- and first-order** models to approximate the behaviour of the system (justify dominant-pole/grouping choices).
- (c) **Plot** the step responses for the original model and your first- and second-order approximations, and decide which is the **lowest-order** version that is usable in practice.

Method

- (a) Decompose the system into its modes and calculate the residuals for each mode.

I'd suggest using the residual formula

$$r_i = \lim_{s \rightarrow p_i} (s - p_i)G(s)$$

for this question. (Or **sympy**, or Wolfram Alpha.)

$$\frac{\frac{1}{22}}{s+1} - \frac{\frac{1}{18}}{s+10} + \frac{\frac{1}{99}}{s+12} = \frac{0.045}{s+1} - \frac{0.055}{s+10} + \frac{0.01}{s+12}$$

$$\frac{0.004}{s+2} + \frac{0.193}{s+5} - \frac{0.197}{s+10}$$

$$\frac{1}{s+5} - \frac{1.33}{s+5.5} + \frac{0.33}{s+7}$$

- (b) Suggest second- and first-order models to approximate the behaviour of the system.

- The poles at -10 and -12 are more than $5\times$ faster than the pole at -1 , so we can remove these, starting with the one at -12 since this is the fastest, and the one with the smallest residual. So:

$$\text{Second-order model: } \frac{1}{(s+1)(s+10)(12)}$$

$$\text{First-order model: } \frac{1}{s+1 \cdot 10 \cdot 12}$$

- Even though the residual of the pole at -2 is very small because of the zero in close proximity, you get a better second-order approximation if you remove the fast pole at -10 . You can then remove the pole at -5 . (I'd suggest plotting the different options so you can see what I mean.)

$$\text{Second-order model: } \frac{s+2.1}{(s+2)(s+5)(10)}$$

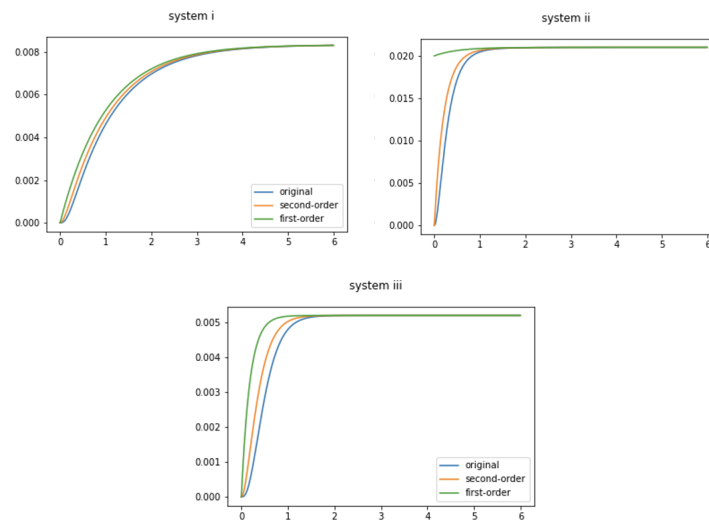
$$\text{First-order model: } \frac{s+2.1}{(s+2) \cdot 5 \cdot 10}$$

- The poles are close together so we'll use the residuals to decide which ones to remove. The pole at -7 has the smallest residual, so that can go first, followed by the residual at -5 . So:

$$\text{Second-order model: } \frac{1}{(s+5)(s+5.5)(7)}$$

$$\text{First-order model: } \frac{1}{s+5.5 \cdot 7 \cdot 5}$$

- (c) Plot the step responses for the original model, and the first- and second-order approximations, and decide which is the lowest-order version that is usable in practice.



The plots above show that:

- Both reduced-order models are close to the original, so we can use the first-order model.
- The second-order model is a good approximation, but the first-order one isn't usable.
- The second-order model is a good approximation, but the first-order one is probably too far from the original for most applications.